

Computability: Exercise 11

Due: June 23, 2026

Question 1

1. Prove that $\text{TIME}(2^n) \not\subseteq \text{TIME}(2^{2n})$.
2. Prove that $\text{NTIME}(n) \not\subseteq \text{PSPACE}$. **Hint:** Use Savitch's theorem.

Question 2

1. Prove that $\text{NP} = \text{PSPACE}$ iff the set of NP-Hard languages is equal to the set of PSPACE-Hard languages.
2. Prove that $\text{co-NP} \subseteq \text{PSPACE}$.

Question 3

Prove that if $\text{P} = \text{NP}$, then $\text{EXP} = \text{NEXP}$ (where $\text{NEXP} := \bigcup_{k=1}^{\infty} \text{NTIME}(2^{n^k})$).

Hint: For a language L , consider the language $L_{\text{pad}} := \{w\#1^{2^{|w|^k}}\}$ (that is, w padded with a $\#$ and $2^{|w|^k}$ ones), for some wisely chosen k .

Question 4

Prove that the language $L = \{\langle M, w, 1^s \rangle : M \text{ is a TM and that accepts } w \text{ using at most } s \text{ space}\}$ is PSPACE-Complete.